Data Analysis - Results

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Part 1

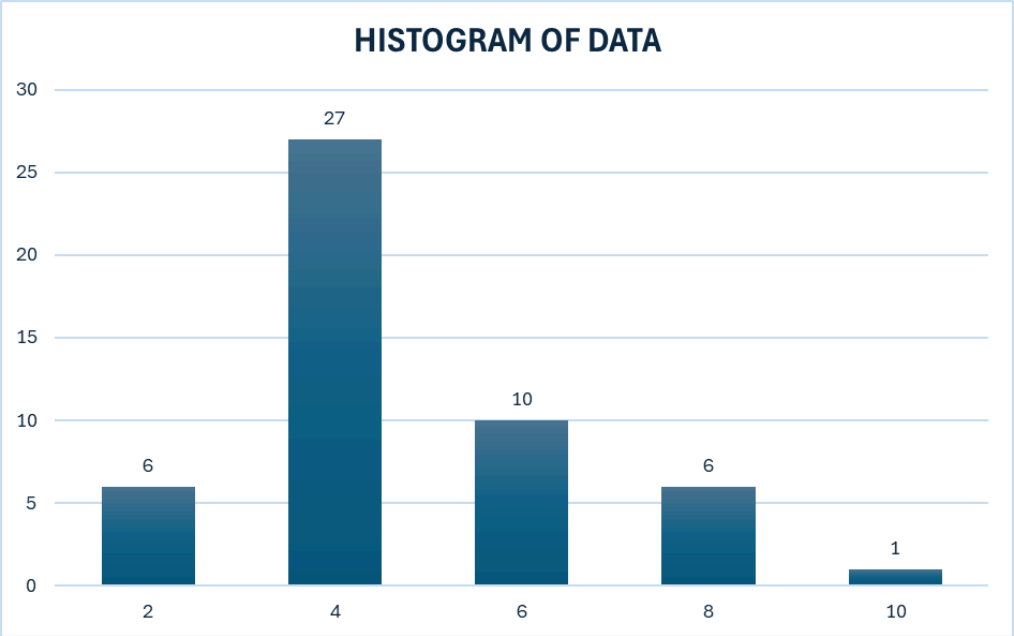
SUMMARY

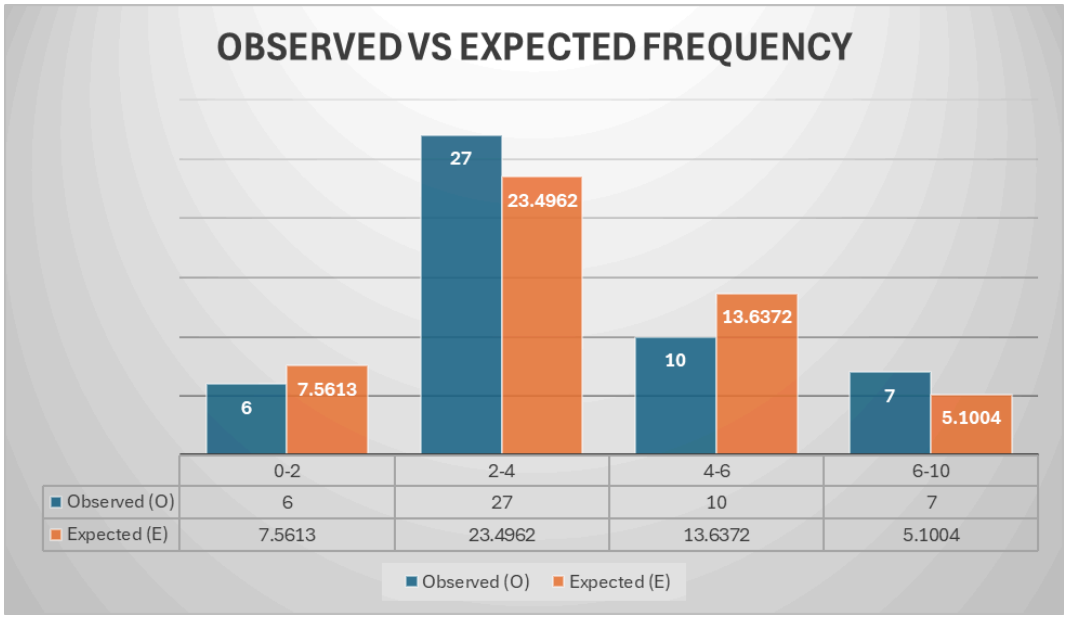
1. Hypothesis & Data Summary
Distribution: Hypothesized Gamma (right-skewed, positive data).
Histogram: Plotted observed frequencies for initial validation.

2. Parameter Estimation (MLE)
Solver Used: Estimated shape (k=3.995) and scale (θ=0.933) by minimizing residual.

3. Goodness-of-Fit Test (Chi-Squared)
Bins: 0–2, 2–4, 4–6, 6–10 (merged for E≥5E≥5).
Observed: Counted using COUNTIFS.
Expected: Probability×50
Result: **Chi-Squared Test**: Compared observed vs. expected frequencies.
Statistic: 2.521
Critical Value: 3.841 (at 95% confidence, 1 degree of freedom).
Since 2.521 < 3.841, the Gamma model **is statistically valid**.

4. Visualization
Histogram of Group 19 AND *Compared Observed vs. Expected frequencies*





***Final Conclusion**: **Gamma distribution is a statistically valid model for the data**.*

Part 2

SUMMARY

Descriptive Statistics

Minimum: 1.0077
Maximum: 1.8118
Mean: 1.2774
Std Dev: 0.1570
Median: 1.2551
Q1: 1.1584
Q3: 1.3759
IQR: 0.2175
Skewness: 0.6904
Mode (approx): 1.2000

Best fitting distribution: beta
Parameters: (np.float64(2.0186253172589326), np.float64(5.8382314303521845), np.float64(1.0041282097256996), np.float64(1.0644040538310202))
SSE: 0.313905

Hypothesis

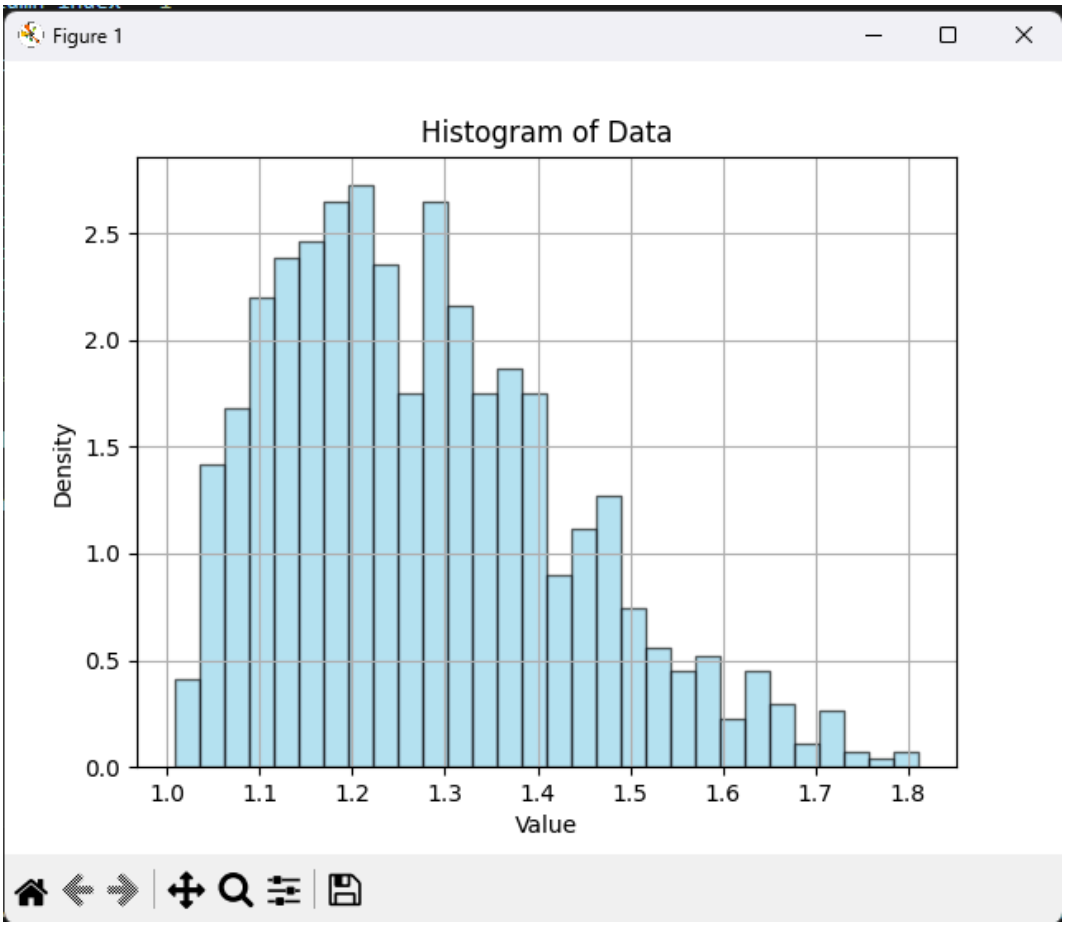
The histogram showed a unimodal, right-skewed distribution bounded between roughly 1 and 2. Based on these traits, a Gamma distribution was initially hypothesized due to its flexibility and suitability for modeling processing times in manufacturing.

Conclusion

Six distribution families were tested: Normal, Exponential, Gamma, Lognormal, Beta and Weibull. Each was fitted using Python and evaluated using the Sum of Squared Errors (SSE) against the histogram's bin densities. Beta provided the best fit, with the lowest SSE of 0.3139. It outperformed the initially hypothesized Gamma distribution, this was confirmed both visually and numerically.

Reliability Discussion

The dataset includes 1,000 values, which provides strong statistical support. Beta distribution closely follows the shape of the empirical data. Given its flexibility to model bounded, skewed distributions, Beta is a statistically reliable and visually appropriate fit for modeling the processing time in this simulation.



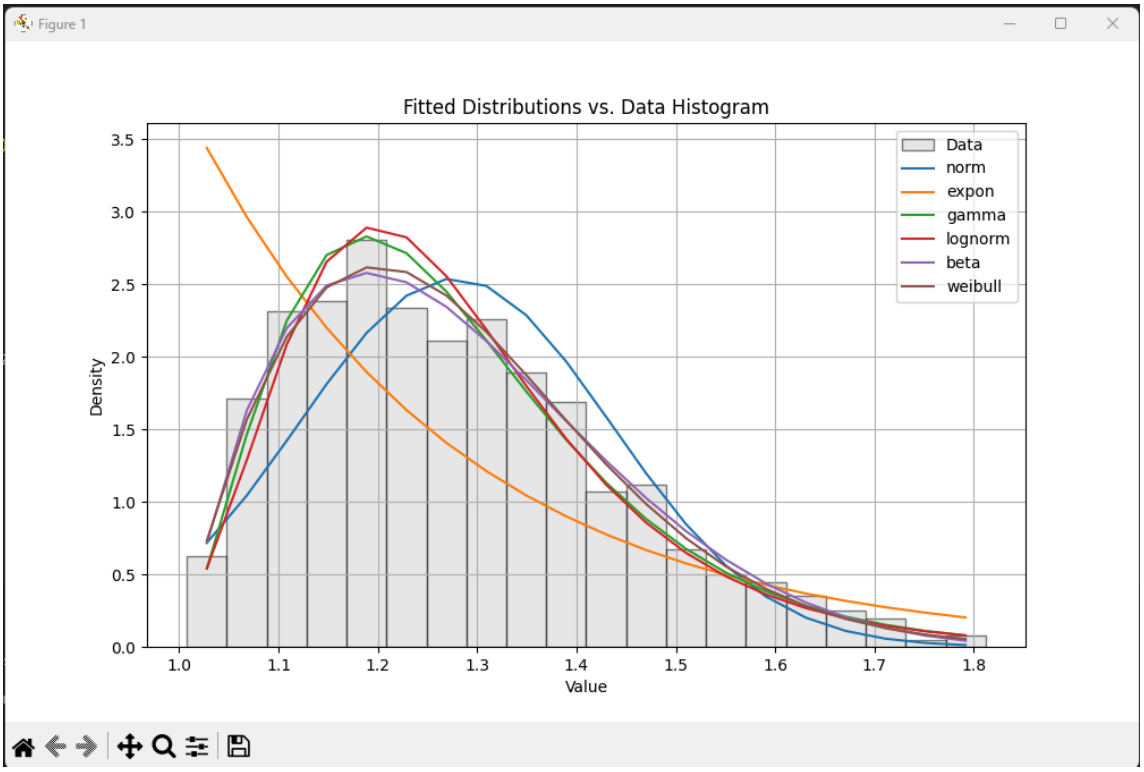


Figure 1
Fitted Distributions vs. Data Histogram
Density
Value
Data
norm
expon
gamma
lognorm
beta
weibull
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8